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August 31, 2022

Dr. Stephanie Johnson
Office of Energy Efficiency and Renewable Energy
Building Technologies Office
U.S. Department of Energy
1000 Independence Avenue SW,
Washington, DC, 20585-0121

Dear Dr. Johnson:

This letter comprises the comments of the Pacific Gas and Electric Company (PG&E), San Diego Gas and Electric (SDG&E), and Southern California Edison (SCE) to show support for the joint proposal submitted on August 23, 2022, to the United States (U.S.) Department of Energy (DOE) detailing scope of coverage, test procedure, and energy conservation standards for consumer room air cleaners.

The signatories of this letter, collectively referred to herein as the California Investor-Owned Utilities (CA IOUs), represent some of the largest utility companies in the Western U.S., serving over 32 million customers. As energy companies, we understand the potential of appliance efficiency standards to cut costs and reduce consumption while maintaining or increasing consumer utility of products. We have a responsibility to our customers to advocate for standards that accurately reflect the climate and conditions of our respective service areas.

The CA IOUs have participated in negotiations with the parties named in the joint proposal submitted on August 23, 2022, from the early stages of negotiations in 2021. We offer our support for the proposal therein and as stated in this letter.

The conclusion of DOE's coverage determination¹ for consumer air cleaners includes whole-home products; however, the joint proposal is limited in scope to consumer *room* air cleaners. Furthermore, consumer room air cleaners represent the majority of sales among these two product classes. The existing U.S. Environmental Protection Agency (EPA) ENERGY STAR® test procedure and coverage for room air cleaners allow DOE to adopt the joint proposal for room air cleaners achieving significant, cost-effective energy savings until a representative test procedure for whole-home products is developed.

The CA IOUs have participated in developing the AHAM AC-7-2022 since March 2022. We recommend that DOE adopt this industry standard as the consumer room air cleaner test procedure, pending any necessary adjustments to ensure that AC-7-2022 represents an average use cycle or period of use as required by the Energy Policy and Conservation Act.

The CA IOUs agree with the joint proposal that using AHAM-7-2022 to implement a two-tiered standard with successive compliance dates is the most appropriate regulatory method to achieve significant energy savings. Listed in Table 1 are the corresponding product classes and the compliance dates for the standard

¹ EERE-2021-BT-DET-0022: <https://www.regulations.gov/document/EERE-2021-BT-DET-0022-0015>; Published July 14, 2022; Final Determination of Air Cleaners.

tiered levels. Please note that the Clean Air Delivery Rate (CADR) product classes are more inclusive than those currently implemented by EPA ENERGY STAR and adopted state standards² because particulate matter 2.5 (PM_{2.5}) is the only particulate of concern accurately measured to 10 CADR.³ The performance metric presented in Table 1 shows an integrated accounting for standby and active mode operation, as detailed in the joint proposal. EPA ENERGY STAR and state standards have independent energy conservation standards for active and standby modes.

Table 1: Product classes for consumer room air cleaners with tiered energy conservation standards and compliance dates.

Product Description	Tier 1	Tier 2
	New Standard (IEF) Dec. 31, 2023	Standard (IEF) Dec. 31, 2025
PM _{2.5} CADR of 10 or greater and less than 100	1.69 PM _{2.5} CADR/Watt	1.89 PM _{2.5} CADR/Watt
PM _{2.5} CADR of 100 or greater and less than 150	1.90 PM _{2.5} CADR/Watt	2.39 PM _{2.5} CADR/Watt
PM _{2.5} CADR of 150 or greater	2.01 PM _{2.5} CADR/Watt	2.91 PM _{2.5} CADR/Watt

Room air cleaners with an active or “automatic mode” use sensors to determine air cleanliness and adjust the fan speed. Operation in automatic mode occurs without human interaction. This technology can potentially save significant energy; however, AHAM AC-1 or AC-7-2022 may not accurately represent these savings. The EPA ENERGY STAR room air cleaner program has not tested automatic mode; therefore, developing a test procedure will require extensive investigation to determine the operation and potential energy savings. We support adopting AHAM AC-7-2022 as the test procedure for room air cleaners to avoid delaying the adoption of room air cleaner energy conservation standards. However, we recommend that DOE ensures the representation of automatic mode operation when it next considers an amended test procedure per 42 U.S.C. 6293(b)(1) and amended energy conservation standards per 42 U.S.C. 6295(m)(1).

² The District of Columbia, Maryland, Nevada, New Jersey, and Washington have passed laws with consumer air cleaner energy conservation standards.

³ As identified in AHAM-AC-1, dust and cigarette smoke particulates (which composes PM_{2.5} in AHAM-AC-7) measure within ten cubic feet per minute and a 95 percent confidence limit (two standard deviations).

In conclusion, we would like to restate our support for the joint proposal submitted on August 23, 2022. We encourage DOE to act expeditiously with a direct final rule to establish the test procedure and energy conservation standards. We thank DOE for the opportunity to be involved in this process.

Sincerely,



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Pacific Gas and Electric Company



Michelle Thomas
Principal Manager, Building Electrification
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Kate Zeng
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